Transport Reviews, 2013
Vol. 33, No. 3, 271–290, http://dx.doi.org/10.1080/01441647.2013.801929

Causes of Youth Licensing Decline: A Synthesis of Evidence

ALEXA DELBOSC* AND GRAHAM CURRIE

Department of Civil Engineering, Institute of Transport Studies, Monash University, Building 69, Clayton, Victoria 3800, Australia

(*Received 27 February 2013; revised 27 March 2013; accepted 30 April 2013*)

ABSTRACT *In recent decades, young adults in many developed nations have become increasingly less likely to acquire a driving license. If this trend continues it could have significant impacts on transport futures. Licensing reductions have only recently been identified and causes are only just being explored. This paper presents a first synthesis of available evidence including an assessment of more influential causal factors. It begins by documenting the declining trend evident in 9 of 14 documented countries; the average rate of decline is 0.6% per annum, with highest declines documented in Australia. A range of causal factors are documented from cross-sectional and longitudinal studies. Changes in life stage and living arrangements, changes in motoring affordability, location and transport, graduated driver licensing schemes, attitudinal influences and the role of e-communication are all explored. Evidence is in general weak and preliminary but suggests multiple causes rather than any single influence. However, of the evidence available life stage factors and affordability influences have stronger links to license decline but are only likely to have a low affect size.*

1. Introduction

Transport in the developed world is heavily dependent on the private car. Although this provides significant mobility benefits, it also results in a range of negative impacts. On-road transport is a significant and increasing source of greenhouse gas emissions (Chapman, 2007). Globally more than one million people are killed and 20–50 million are injured by road collisions each year (Nantulya et al., 2003). It is now generally recognised that a sustainable future requires a rethink of this dependence on the private car (Toleman & Rose, 2008). The historical growth in car travel and the embedded stability of the dominant socio-technical regimes that support automobile-based transport are seen as barriers to achieving this aim (Geels, 2012).

Against this backdrop a significant opportunity is arising: in many countries young people are less likely to get a car license and, if they can drive, they are driving less (Kuhnimhof, Buehler, & Dargay, 2011; Raimond & Milthorpe, 2010; Sivak & Schoettle, 2011, 2012a). This trend has only recently been identified and researchers are just beginning to explore its causes.

*Corresponding author. Email: alexa.delbosc@monash.edu

Although acquiring a license does not automatically mean someone will drive a car, it is a necessary first step towards car-based mobility. This decline in youth licensing is an important trend to understand; generation Y recently surpassed the baby boomers to become the largest generation in both the USA and Australia (Australian Bureau of Statistics, 2011b; Lachman & Brett, 2011); so understanding their travel needs will become increasingly important for forecasting travel demand and mode choice. As noted in this issue, several industrialised nations have identified stagnating or decreasing car travel demand (Headicar, 2013; Kuhnimhof, Buehler, Wirtz, & Kalinowska, 2012b; Le Vine, Jones, & Polak, 2009; Metz, 2013; Newman & Kenworthy, 2011; Puentes & Tomer, 2008; Zumkeller, Chlond, & Manz, 2004) and some researchers are already estimating the extent to which changes in licensing are contributing to this trend (Kuhnimhof, Zumkeller, & Chlond, 2013a, 2013b).

This paper provides a review of the evidence of car licensing decline among young people in the developed world. It also reviews evidence of causal factors and weighs the likely impacts of these influences. The paper commences with a review of evidence of declining licensing rates. This is followed by a review of evidence regarding causes. A synthesis of this evidence is then discussed in the conclusion.

2. The Decline in Youth Driver Licensing

The decline in driver licensing among young people was first recognised in Sweden and Norway in the late 1990s, where licensing of young adults dropped by over 10% between the mid-1980s and the late 1990s (Berg, 2001). The research that followed was limited primarily to Scandinavia (e.g. Ruud & Nordbakke, 2005) and did not gain wider traction, perhaps because other countries were not yet experiencing this trend; Noble (2005) is one exception of early research in the UK. This changed by around 2010 when researchers began to recognise a drop in licensing in Australia, North America, Japan and much of Europe.

The trends within developed countries are presented in Figure 1 and detailed in Table 1. The figure shows available published trend data for young people aged between 18 and 30 for 14 countries with at least five years between survey periods. These data are usually sourced from national household travel surveys although some are taken directly from licensing authorities. In Australia where no national travel survey exists, data are drawn from state travel surveys and licensing authorities.

Figure 1 illustrates that in 9 of 14 countries, youth licensing rates are in decline. Some of the fastest changes have occurred in two states of Australia. In the states of Victoria and New South Wales the per cent of young adults with a driving license dropped between 0.9% and 1.0% per year (Australian Bureau of Statistics, 2010; Raimond & Milthorpe, 2010; VicRoads, 2012).

In the USA between 1983 and 2010, licensing dropped for all age groups below 30 with the largest decreases among the youngest age groups (Kuhnimhof et al., 2012a; Sivak & Schoettle, 2012b). In Canada between 1999 and 2009 the trend is not as evident with only a marginal drop among those aged under 25 but a more noticeable drop for those aged 25–34 from over 90% to around 85% (Sivak & Schoettle, 2012a).

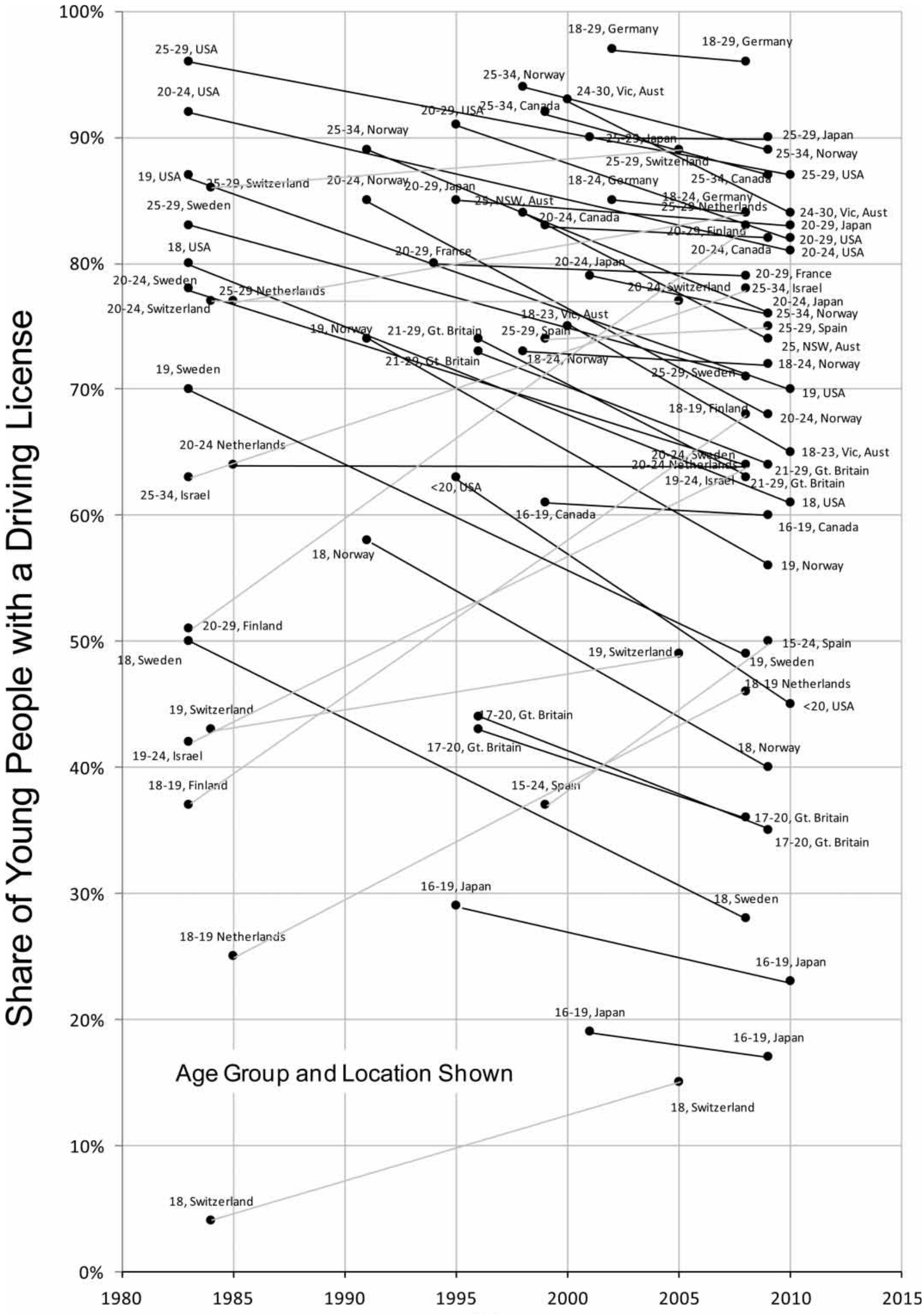

Figure 1. Young people driver licensing trends from international data.
Source: Raimond and Milthorpe (2010), Kuhnimhof et al. (2012a), Sivak and Schoettle (2012a, 2012b), and Delbosc and Currie (2013).

Reviews by Sivak and Schoettle (2012a) and Kuhnimhof et al. (2012a) showed that in Europe the trend is somewhat mixed. Supporting the earliest research in licensing, the strongest declines in Europe have occurred in the Scandinavian

Table 1. Trends in driver licensing of young people in 13 developed countries

Location	Years	Age group (years)	Licensing change (%)	Change per year (%)	Source
Victoria, Australia	2000/01–2010/11	18–23	75–65	−1.0	Author's analysis (Australian Bureau of Statistics, 2010; VicRoads, 2012)
		24–30	93–84	−0.9	
NSW, Australia	1998–2009	25	84–74	−0.9	Raimond and Milthorpe (2010)
USA	1983–2010	18	80–61	−0.7	Sivak and Schoettle (2012b)
		19	87–70	−0.6	
		20–24	92–81	−0.4	
		25–29	96–87	−0.3	
	1995–2010	<20	63–45[a]	−1.2	Kuhnimhof et al. (2012a)
		20–29	91–82[a]	−0.6	
Norway	1991–2009	18	58–40	−1.0	Sivak and Schoettle (2012a)
		19	74–56	−1.0	
		20–24	85–68	−0.9	
		25–34	89–76	−0.7	
	1998–2009	18–24	73–72[a]	−0.1	Kuhnimhof et al. (2012a)
		25–34	94–89[a]	−0.5	
Sweden	1983–2008	18	50–28	−0.9	Sivak and Schoettle (2012a)
		19	70–49	−0.8	
		20–24	78–64	−0.6	
		25–29	83–71	−0.5	
Great Britain	1995/97–2008	17–20	43–36	−0.5	Sivak and Schoettle (2012a)
		21–29	74–64	−0.8	
	1995/97–2009	17–20	44–35[a]	−0.7	Kuhnimhof et al. (2012a)
		21–29	73–64[a]	−0.7	
Japan	2001–2009	16–19	19–17	−0.3	Sivak and Schoettle (2012a)
		20–24	79–76	−0.4	
		25–29	90–90	0.0	
	1995–2010	16–19	29–23[a]	−0.4	Kuhnimhof et al. (2012a)
		20–29	85–83[a]	−0.1	
Germany	2002–08	18–24	85–84	−0.2	Sivak and Schoettle (2012a)
	2002–08	18–29	97–96[a]	−0.2	Kuhnimhof et al. (2012a)
France	1994–2008	20–29	80–79[a]	−0.1	Kuhnimhof et al. (2012a)
Canada	1999–2009	16–19	61–60	−0.1	Sivak and Schoettle (2012a)
		20–24	83–82	−0.1	
		25–34	92–87	−0.5	
Finland	1983–2008	18–19	37–68	+1.2	Sivak and Schoettle (2012a)
		20–29	52–83	+1.2	
Israel	1983–2008	19–24	42–64	+0.9	Sivak and Schoettle (2012a)
		25–34	63–78	+0.6	
The Netherlands	1985–2008	18–19	25–46	+0.9	Sivak and Schoettle (2012a)
		20–24	64–64	0.0	
		25–29	77–84	+0.3	

(Continued)

Table 1. Continued

Location	Years	Age group (years)	Licensing change (%)	Change per year (%)	Source
Switzerland	1984–2005	18	4–15	+0.5	Sivak and Schoettle (2012a)
		19	43–49	+0.3	
		20–24	77–77	0.0	
		25–29	86–89	+0.1	
Spain	1999–2009	15–24	37–50	+1.3	Sivak and Schoettle (2012a)
		25–29	74–75	+0.1	

[a]Percentages are taken from graphs and should be taken as approximate.

countries of Norway and Sweden. Great Britain also saw significant decreases in licensing between 1995 and 2008, especially for those aged 21–29. In Germany licensing has remained relatively stable; however other research suggests that young Germans are becoming less likely to have a car and are more likely to use other travel modes (Kuhnimhof et al., 2011, 2012b).

Most Asian countries are still economically developing and currently have very low levels of household car ownership; countries such as India and China are projected to rapidly increase their motorisation in the coming decades (Chamon, Mauro, & Okawa, 2008). However, in Japan, a country with relatively high motorisation, youth driver licensing decreased slightly between 2001 and 2009 from 79% to 75% of those aged 20–24 (Sivak & Schoettle, 2012a).

The data in Figure 1 also show that in five countries youth licensing rates are increasing, in some cases quite dramatically. Finland has seen an increase in licensing between 1983 and 2008 from 51% to 82% of 20–29-year-olds. Israel, the Netherlands, Switzerland, Spain and Latvia all showed at least modest increases in licensing (Sivak & Schoettle, 2012a). However, it is worth noting that increases have generally occurred in countries where licensing rates were at the lower end of the scale. In addition, in each case, only two survey years of data are available and in some cases they are over 20 years apart. It is possible, therefore, that in some countries licensing peaked in the 1990s and have begun to decline, but not to levels below those in the 1980s. For example, long-term licensing data from the UK show that youth licensing peaked in the mid-1990s (Metz, 2012; National Travel Survey, 2010); if only data from 1985 and 2010 were compared then the UK would appear to have *increased* licensing rates. However, this is not the case for the Netherlands, where youth licensing continued to increase between 1995 and 2009 (van der Waard, Jorritsma, & Immers, 2013).

It is difficult to aggregate these trends as sample sizes will vary greatly. However, an unweighted average suggests that overall licensing has declined by 0.2% per year. For those countries showing a decline in licensing, licensing rates declined on average by 0.6% per year but this ranges between 0.1% and 1.0% per annum. Overall, declines were not consistently more pronounced for any particular age group; however, there is some suggestion that the rate of decline is higher in more recent years.

In summary, in many developed countries people under the age of 30 are less likely to hold a driving license than they were 10–20 years earlier. Among those countries where licensing is decreasing, the average annual rate of decrease is

−0.6% per year with most values between −0.3% and −1.0%. It is worth noting that a recent review found that in many countries licensing declines are more pronounced among young men than among young women (Kuhnimhof et al., 2012a).

The following sections of this paper will focus on those countries where licensing is declining in an attempt to understand causes.

3. Why Are Young People Less Likely to Get a License?

Because the trend towards reduced youth licensing has only recently been noted, research into why this is occurring is still emerging. No clear overall picture of causal factors has been established — hence the need to consolidate available evidence.

Table 2 presents a synthesis of hypothesised causal factors suggested in the research literature. Some studies directly measured both license-holding and potential causes in the same study, although these studies are generally cross-sectional in nature and so they cannot directly measure trends. Other evidence is indirect, that is, it identifies a long-term change in a potential cause but does not directly measure both licensing rates and the hypothesised cause in the same study.

Table 2 classifies potential causal factors into six categories:

(1) Life stage,
(2) Affordability
(3) Location and Transport,
(4) Driver licensing regulations,
(5) Attitudes
(6) E-communication

The following sections summarise available evidence presented in each of these categories.

3.1 *Life Stage*

Some research examines how household variables influence licensing among young people. Two studies have found that if a young person is a student or working part time they are less likely to have a license than if they have a full-time job; for example, in the UK 58% of young adults with a full-time job had a license compared to 38% with a part-time job and 32% of students (Delbosc & Currie, 2013; Noble, 2005). Young adults who are living independently with their own children are more likely to hold a license (Delbosc & Currie, 2013); however, the effect of living in the parental home is inconsistent (Licaj, Haddak, Pochet, and Chiron, 2012). Young adults in the parental home are likely to have access to shared household vehicles and the lower housing costs can free up income that could be spent on a car; however, living with parents may indicate that a young adult is financially dependent and unable to pay for a car.

Indirect, longitudinal research shows that the above demographic influences have been changing in recent decades in many countries which could act to influence licensing rates. Rates of educational participation have increased and full-time employment has decreased among young adults, both of which are likely to reduce the financial ability to purchase and run a car in the

Table 2. Hypothesised reasons for decreased licensing

Explanation	Direct evidence	Indirect evidence	Notes
Life stage			
Increasing rate of educational participation	Noble (2005) and Delbosc and Currie (2013)	Australian Bureau of Statistics (2011c), Office for National Statistics (2011), Taylor et al. (2012), and van der Waard et al. (2013)	Licensing also dropping among non-students (Noble, 2005)
Decreasing employment rates	Noble (2005) and Delbosc and Currie (2013)	Bureau of Labor Statistics (2011), Office for National Statistics (2011), Australian Bureau of Statistics (2012b), and Taylor et al. (2012)	
Delaying marriage/ children	Delbosc and Currie (2013)	Mitchell (2006), Australian Bureau of Statistics (2012a), and Stokes (2012)	
Living with parents longer	Licaj, Haddak, Pochet, and Chiron (2012) and Delbosc and Currie (2013)	Mitchell (2006), Cobb-Clark (2008), Settersten and Ray (2010), Office for National Statistics (2012), Stokes (2012), and Taylor et al. (2012)	Unclear whether living with parents increases or decreases likelihood of licensing
Affordability			
Insurance cost	Noble (2005)		Very rapid increase in the UK
Cost of license	Noble (2005)		Significant increase in the UK
Cost of petrol		Davis, Dutzik, and Baxandall (2012)	
Cost of car purchase	Noble (2005) and KRC Research (2010)		Real cost of car purchase declining (Noble, 2005)
Costs — general	Berg (2001) and Williams (2011)		
Household income	Licaj et al. (2012) and Delbosc and Currie (2013)		Biggest drop in miles driven is among young men of higher income (Stokes, 2012)
Recession/economy		Berg (2001) and Davis et al. (2012)	Trend began before recession (Davis et al., 2012; Newman & Kenworthy, 2011); drop continued in Sweden after 1990 recession ended (Berg, 2001)
Location and Transport			
Use PT/other modes instead	Berg (2001), Noble (2005), and Williams (2011)	Raimond and Milthorpe (2010), Davis et al. (2012), Kuhnimhof et al. (2012a), and van der Waard et al. (2013)	

(*Continued*)

Table 2. Continued

Explanation	Direct evidence	Indirect evidence	Notes
Moving to inner-city/accessible areas	Noble (2005), McDonald and Trowbridge (2009), Raimond and Milthorpe (2010), and Licaj et al. (2012)	Environmental Protection Agency (2010), Raimond and Milthorpe (2010), Belden Russonello and Stewart (2011), Lachman and Brett (2011), Davis et al. (2012), and van der Waard et al. (2013)	Car travel/licensing also declining in non-urban areas (Grimal, Collett, & Madre, 2013; Noble, 2005; van der Waard et al., 2013)
Driver Licensing Regulations			
Licensing regimes became more strict	Noble (2005), Raimond and Milthorpe (2010), and Williams (2011)	Preusser and Tison (2007) and Senserrick (2009)	Began dropping before GDL in Australia (Raimond & Milthorpe, 2010)
Household car access/driving supervisor	Williams (2011), Licaj et al. (2012), and Delbosc and Currie (2013)		
Attitudes			
Want to help the environment	Forward, Åretun, Engström, Nolén, and Börjesson (2010), and KRC Research (2010)	Department for Environment Food and Rural Affairs (2002) and Noble (2005)	Most evidence suggests this is not influential in license decline (Delbosc & Currie, 2012)
Cars no longer a status symbol	Berg (2001)	Steg (2005), Kalmbach, Bernhart, Kleimann, and Hoffman (2011), Delbosc and Currie (2012), and van der Waard et al. (2013)	Some evidence suggests that young adults *more* likely to see car as status symbol
Too busy/other priorities	Berg (2001), Noble (2005) and Williams (2011)		
E-communication			
E-communications replacing face-to-face contact	KRC Research (2010) and Williams (2011)	Sivak and Schoettle (2012a)	Some evidence suggests this is not influential in license decline (Delbosc & Currie, 2012)
E-communications suit PT use		Davis et al. (2012)	

short term. For example, in Australia, the per cent of people aged 20–24 attending some form of education increased from 25% in 1991 to 41% in 2011 (Australian Bureau of Statistics, 2011c). In the USA, enrolment in high school or university increased from 31% in 1990 to 45% in 2011 (Taylor et al., 2012); similarly, attendance of summer school (traditionally a time for summer employment) has steadily grown since the mid-1990s (Bureau of Labor Statistics, 2011). In the UK, full-time educational enrolment increased from 10% in 1992 to almost 20% in 2011 (Office for National Statistics, 2011). In the Netherlands, for those aged 18–25 participation in higher education has increased 40% between 2001 and 2011 (van der Waard et al., 2013).

In the USA, the employment−population ratio for young adults has decreased from a high of 67% in 1990 to 49% in 2010 (Bureau of Labor Statistics, 2011). Figure 2 shows that the proportion of young Australians in full-time work has dropped from almost 50% in 1985 to 33% in 2011 whereas part-time work has steadily increased.

Young adults have been living at home for longer in Australia, North America and Europe (Australian Bureau of Statistics, 2009; Cobb-Clark, 2008; Mitchell, 2006; Settersten & Ray, 2010). For instance, the per cent of Canadians aged 20–29 living in the parental home jumped from 27% in 1981 to 41% in 2001 (Mitchell, 2006). Almost half of young Australians who had not left home by 24 said their main reason was to save money (Australian Bureau of Statistics, 2009). Parents appear to recognise this shift, as the per cent of US parents who thought that children should be financially independent by the time they were 22 dropped from 80% in 1993 to 67% in 2011 (Taylor et al., 2012).

Similarly, young adults are marrying and having children later in life; for example, the median age of first marriage in Australia has increased over three years since 1990 (Australian Bureau of Statistics, 2012a; Mitchell, 2006) and young Australians are much less likely to live with a spouse or partner than they were in the 1980s (Australian Bureau of Statistics, 2009).

Taken together, these influences may act to delay the traditional 'markers' that symbolise a transition to adulthood. For many young people, this may result in a postponement of the need for and the financial ability to pay for a car. However, affordability is a complex issue; young adults living in the parental home are likely to have access to parental resources such as low housing costs and shared household vehicles. This makes it difficult to discern whether the financial resources of young adults have changed in recent years. The next section examines whether the cost of motoring could be an influence on the change in licensing.

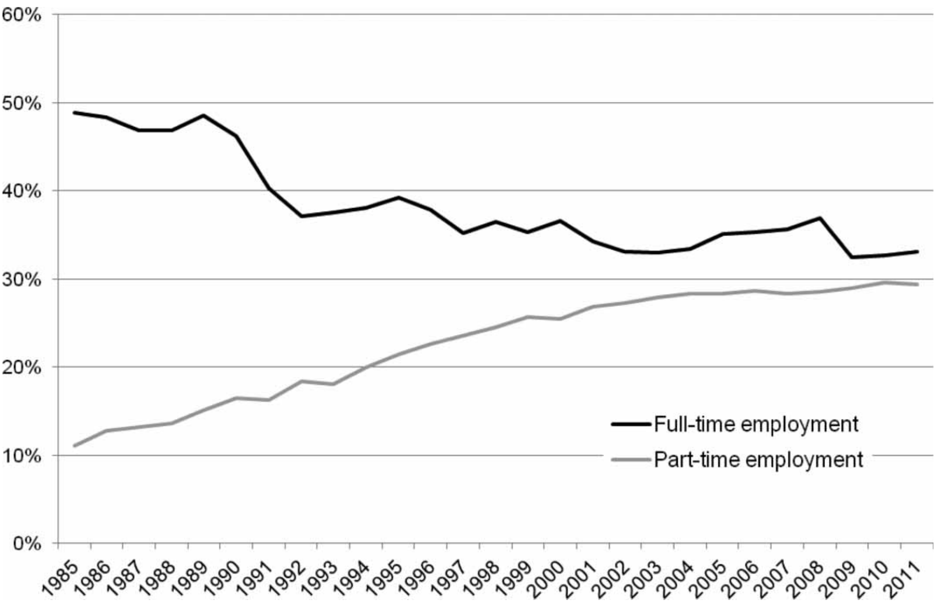

Figure 2. Per cent of Australians aged 15–24 in employment, 1985–2011.
Note: Employment data seasonally adjusted. Population and employment taken at June 30.
Source: Australian Bureau of Statistics (2011d, 2012b).

3.2 *Affordability*

Some researchers have explored the role of the costs of motoring on youth licensing trends. Household income has been identified as a significant influence on whether a young person holds a driver's license (Delbosc & Currie, 2013; Licaj et al., 2012). It is not surprising, then, that financial shocks like major recessions have been proposed as a major factor in reduced youth licensing (Berg, 2001; Davis et al., 2012). For example, the global financial crisis of 2008 and the resulting increase in unemployment rates had a disproportionate impact on young people (Davis et al., 2012).

However, it is not clear whether short-term financial shocks are a significant influence on licensing. In Sweden, for example, licensing rates began to fall before a recession in the 1990s and continued to fall afterwards (Berg, 2001). More recently in the USA, even young people with jobs or in high-income households are more likely to use public transport and less likely to drive than a decade ago (Davis et al., 2012). In both the USA and Australia, vehicle miles travelled began to drop several years *before* the global financial crisis (Davis et al., 2012; Newman & Kenworthy, 2011). Furthermore, an analysis in the UK found the largest drop in miles driven since 1995 was among young men of higher incomes and men in full-time work (Stokes, 2012).

Similarly, it is unclear whether the costs of owning and running a car have a significant influence on the licensing trend. In some cases costs have rapidly increased; for example, in the UK the cost of acquiring a license has increased significantly and insurance now makes up over half the cost of motoring for a male teenager (Noble, 2005). Similarly, in many countries petrol prices have increased significantly in recent decades and are unlikely to significantly reduce (Davis et al., 2012). Overall, however, the real cost of motoring has declined compared to inflation, driven primarily by reduced purchase costs (Noble, 2005).

The actual cost of motoring, however, may not be as influential as the *perceived* cost. Several surveys have found that young people cite costs as a reason for not getting a license (Berg, 2001; KRC Research, 2010; Noble, 2005; Williams, 2011). For example, 30% of US 18-year-olds who had not yet begun to get a license said this was because it 'cost too much', although this explanation was fourth behind 'no car available', 'can get where want without driving' and 'too busy' (Williams, 2011). In the UK, on the other hand, the top three main reasons for not driving were the cost of learning to drive, the cost of insurance and the cost of buying a car (Noble, 2005). However, without longitudinal data it is unclear whether this has changed over time.

It is likely that income and costs have at least some impact on the change in youth licensing, although it is clear that these influences are complex. It may be that other influences are causing some young people to reprioritise their spending away from cars and towards other necessities and interests. An analysis of the Australian Household Expenditure Survey, for example, suggests that young households are spending a significantly larger proportion of their income on housing compared to a decade ago (Figure 3). Discretionary spending on clothing, alcohol and recreation does not appear to have reduced but spending on transport is slightly lower (Australian Bureau of Statistics, 2011a).

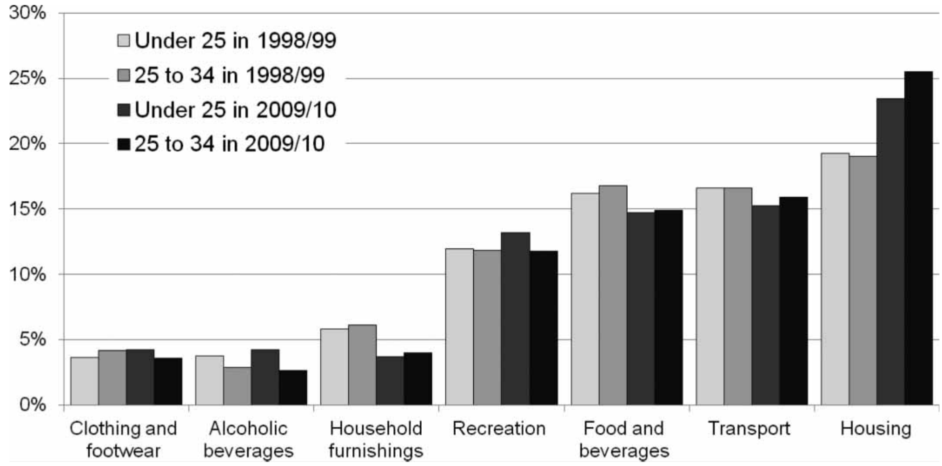

Figure 3. Australian household expenditure by age, 1998/99 and 2009/10.
Note: Shown as percentages of total weekly expenditure. Young adults living in parental home are not included.
Source: Australian Bureau of Statistics (2000, 2011a).

3.3 *Location and Transport*

Accessible and inner-urban areas facilitate living without a car and several researchers have examined the relationship between household location and youth licensing. In the USA and Australia, residential building approvals in inner-city accessible areas is rapidly increasing (Environmental Protection Agency, 2010; Raimond & Milthorpe, 2010). Young adults, especially those without children, are a major market for this trend, with many preferring to live in mixed-use and walkable 'smart growth' suburbs or inner cities (Belden Russonello & Stewart, 2011; Davis et al., 2012; Lachman & Brett, 2011). A similar trend of urbanisation has been observed in the Netherlands (van der Waard et al., 2013).

Young people living in inner and accessible areas are less likely to have a driving license compared to those living in outer-urban or rural areas (Licaj et al., 2012; McDonald & Trowbridge, 2009; Noble, 2005; Raimond & Milthorpe, 2010), although it is important to note that in the UK, France and the Netherlands, licensing and car travel is also decreasing in small-town and rural areas (Grimal et al., 2013; Noble, 2005; van der Waard et al., 2013). The direction of causality is unclear — young adults who are attracted to inner-city living may choose not to get a license or young adults who do not want a license end up living in inner-city areas. It is reasonable to assume that causality works both ways.

Young people without a license must use alternative transport including lifts, public transport or living in areas where walking and cycling can meet travel needs. Several surveys have found that being able to 'get around without driving' is a common reason given for not having a license (Berg, 2001; Noble, 2005; Williams, 2011). Indirect research has examined which alternative travel mode is being adopted. In the USA between 2001 and 2009, young people increased cycling by 24%, walking by 16% and public transport trips by 40% (Davis et al., 2012). In Sydney, between 1991 and 2008 young people increased their travel as a car passenger but did not increase public transport, walking or cycling in that time (Raimond & Milthorpe, 2010). An analysis of household travel surveys in Victoria, Australia, found that the per cent of people aged 17–34 driving on the day of the survey decreased significantly whereas car passenger trips dropped for those under 20 (see Figure 4). Public transport use decreased for those under 20 but increased for those aged 21–40. Walking decreased for all age groups across the survey period. In the Netherlands, young adults are less likely to use the car for travel and are making fewer trips overall between 1995 and 2009 (van der Waard et al., 2013).

3.4 *Driver Licensing Regulations*

Since the 1990s, many developed nations have introduced graduated driver licensing (GDL) schemes which expand training requirements for learner and 'provisional' drivers before the transition to a full driving license. The UK is a notable exception to this, although the licensing test became significantly more expensive and difficult in 2003 (Noble, 2005).

Although GDL schemes vary between countries (and between states within countries), common characteristics include increasing the minimum age for a permit, mandatory driving lessons, stricter theory and practical testing, minimum supervised driving hours, curfews, maximum speed or engine size restrictions, blood alcohol restrictions, mobile phone restrictions or restrictions on the number of non-relative passengers (Preusser & Tison, 2007; Senserrick, 2009). In some jurisdictions some of these restrictions are waived once people reach a certain age (Senserrick, 2009).

The purpose of these restrictions is to improve road safety and numerous evaluations suggest that GDL achieves this goal (Preusser & Tison, 2007). However, another possible consequence of GDL is that it acts as a barrier to accessing a license. Raimond and Millthorpe (2010) have found that an increasing number of young people are not converting a learner's permit to a provisional license. Common reasons for not having a license among US teenagers included "licensing requirements a hassle" and "parents too busy to supervise [driving]" (Williams, 2011). The requirement for a minimum number of supervised driving hours (20–60 hours in US states and up to 120 hours in some Australian states) may be particularly off-putting, especially for young adults without access to parental cars. The presence of household vehicles is already one of the strongest predictors of youth licensing (Delbosc & Currie, 2013; Licaj et al., 2012).

However, other evidence suggests that GDL has only a minor impact on overall licensing rates. In Australia, licensing rates began to drop before GDL was introduced; similarly licensing rates in the UK declined before the stricter test was introduced in 2003 (Noble, 2005). Furthermore, reductions identified in the USA appear to 'rebound' quickly (Zhu, Chu, & Li, 2009). In addition, the UK has no graduated licensing — learner drivers can become full drivers as long as they pass a theory test and a practical driving test (Directgov, 2012) — yet, there, too, youth licensing rates are declining. It may be that GDL is not onerous enough to keep young adults away from a driver license if they really need one, yet difficult enough to provide one more reason not to get one if it is not needed.

3.5 *Attitudinal Influences*

In addition to demographic and structural changes, attitudes may play a role in reducing the attractiveness of the car as a mode of transport. Some researchers

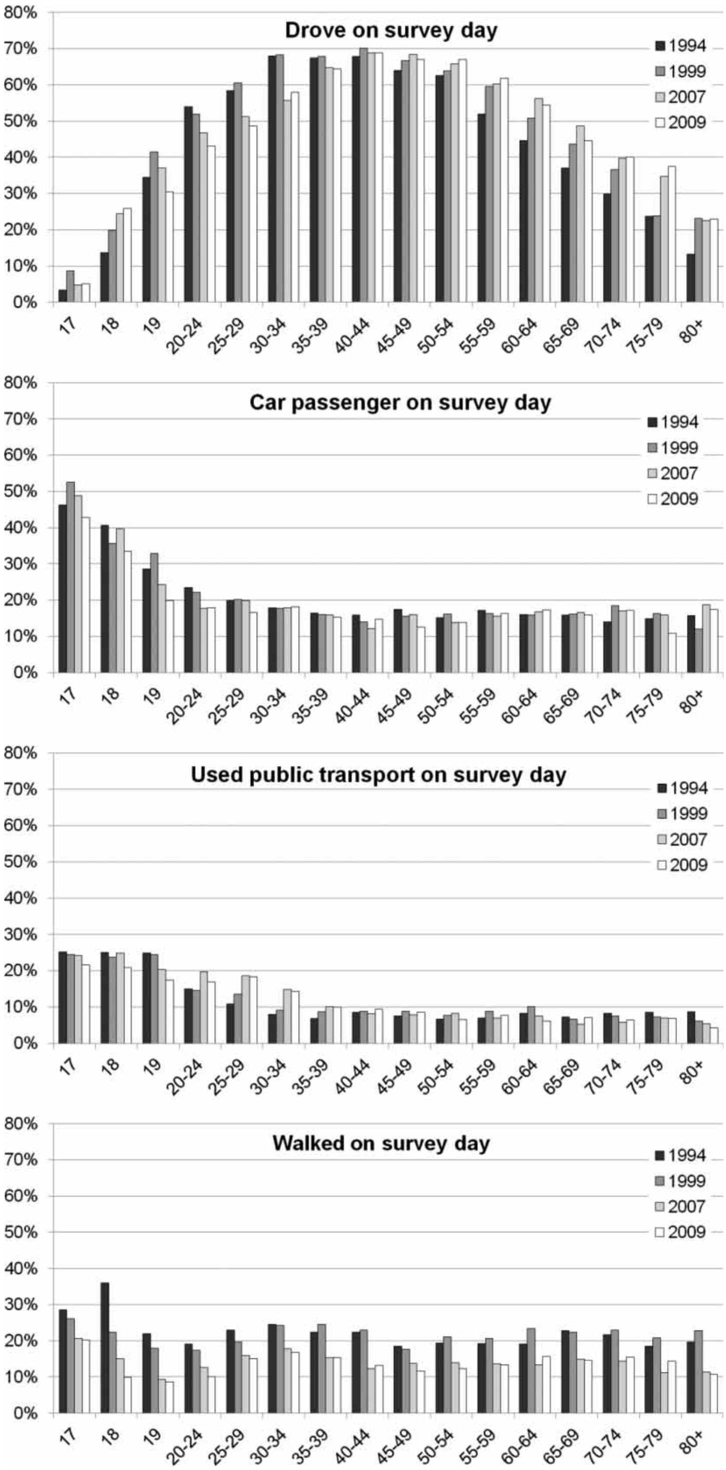

Figure 4. Mode use on day of travel survey by age, Victoria, 1994–2009.
Note: Minimum legal age of unsupervised driving in Victoria is 18.
Source: Author's analysis of VATS and VISTA travel surveys (Department of Transport, 2009; The Transport Research Centre, 2001).

have suggested that environmental attitudes may be discouraging young adults from driving cars. An online survey of 300 young adults commissioned by the car-sharing company Zipcar found that almost half of the young adults agreed with the statement "I want to protect the environment, so I drive less" (KRC Research, 2010) and a Swedish longitudinal survey did find a slight increase in the number of young people citing environmental reasons as a reason not to get a license (Forward et al., 2010).

However, other research casts doubt on whether young people are more environmentally aware than previous generations and questions whether these attitudes have an impact on behaviour. A large-scale survey in the UK found that people aged 18–24 were the least likely of any age group to recycle, cut down on energy and water use, or say they cut down on car use to save the environment (Department for Environment Food and Rural Affairs, 2002). A qualitative study in Australia found that environmental issues were not top-of-mind for young people and many downplayed any possible impact their travel could have on the environment (Delbosc & Currie, 2012). These studies suggest that environmental attitudes are unlikely to be a major contributor to these trends.

Cars have long been seen as a symbol of freedom, independence and social status (Steg, 2005; Stokes & Hallett, 1992), and young adults who hold these views are more likely to have a license (Berg, 2001). However, there is some suggestion that the social status of the car may be waning for younger generations. Qualitative research conducted in Australia suggests that the car has become a symbol of adult responsibility rather than an aspirational symbol of status (Delbosc & Currie, 2012). In Europe and North America, some young adults are prioritising other pursuits over getting a car license; being 'too busy' to get a license has been cited in several studies (Berg, 2001; Noble, 2005; Williams, 2011). However, this trend is by no means clear; research in the Netherlands found that young people were just as likely as their parents to believe that a car gives them prestige and for many owning a car is still a strong aspiration (Steg, 2005; van der Waard et al., 2013).

If the car is losing its place as a major interest and status symbol, it is possible that its position is being displaced by technology and e-communications. A survey in Japan found that 27% of people aged 40–59 listed cars as an interest, ranking seventh (just ahead of computers); in contrast, this dropped to 23% of those aged 18–24, ranking 17th behind computers, portable music players, anime, video games and TV. The following section will explore the impact of e-communications and car licensing more specifically.

3.6 *The Role of e-Communication*

Understanding the interaction between transport and e-communication is a complex and fast-changing topic. There is certainly no doubt that young adults spend considerable time and money on computers, mobile phones and the internet. Figure 5 demonstrates the different household expenditure patterns for selected communication equipment in Australia. Young adult households spend a far greater proportion of their income on mobile phone service than other household ages and slightly higher amounts on computers and internet charges (Australian Bureau of Statistics, 2011a).

Because of these differences, much of the media speculation about the cause of licensing declines leaps to the assumption that electronic communication is

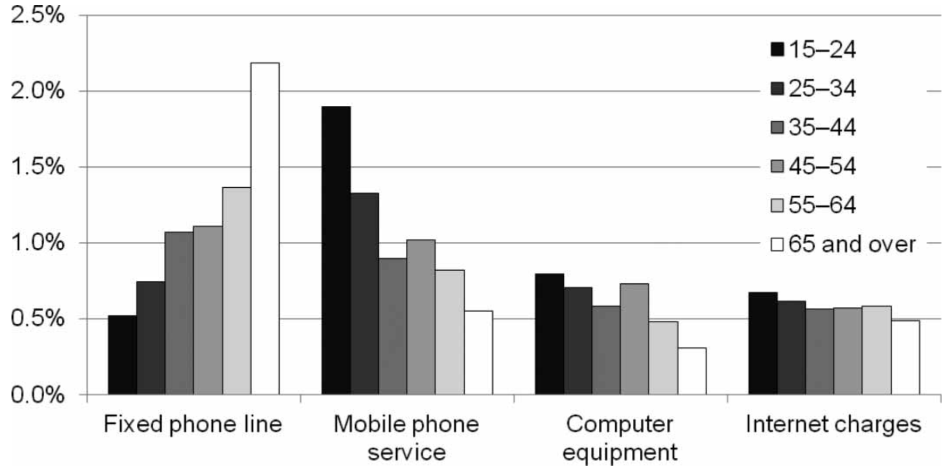

Figure 5. Australian household expenditure by age, 2009/10.
Note: Shown as percentages of total weekly expenditure. Young adults living in parental home are not included.
Source: Australian Bureau of Statistics (2011a).

reducing the need for young people to travel, with conclusions such as "today Facebook, Twitter and text messaging allow teenagers and 20-somethings to connect without wheels" (Chozick, 2012, p. 1).

This effect is supported by an online questionnaire by Zipcar that found that 50% of young people say they sometimes spend time with friends online instead of in person, and an analysis by Sivak and Schoettle (2012a) that found a relationship between a country's internet penetration rate and youth licensing. Up to 10% of US teenagers said they did not have a license because e-communications keep them in touch with friends, although no respondents cited this as their primary reason (Williams, 2011).

However, other research casts doubt on these conclusions. Some researchers question the validity of the analysis used to establish a relationship between national internet penetration rates and youth licensing (Le Vine, Latinopoulos, & Polak, 2013; Sivak & Schoettle, 2013). Qualitative research in Australia suggests that e-communication supplements face-to-face contact but cannot replace it (Delbosc & Currie, 2012). An analysis of social networks in the Netherlands found that the more frequently people contacted others by telephone, e-mail or texting, the more frequently they maintained face-to-face contact as well (van den Berg, Arentze, & Timmermans, 2009).

Furthermore, these studies tend to share the flawed assumption that replacing the need to *travel* equates with reducing the need to *drive.* If e-communications reduce the need for face-to-face contact, this should reduce all travel, not just car trips.

However, there is at least one other way that electronic communications could influence licensing. E-communication facilitates a lifestyle based around public transport travel better than car travel (Davis et al., 2012). While travelling on a train or bus, young adults can maintain contact through phone calls, email, texting or updating their status on social media websites. Furthermore, e-communications are making public transport travel easier as more transit systems provide real-time information and network updates through websites and mobile apps. In contrast, most countries and states ban the use of mobile phones or texting whilst driving, cutting off young people from communication while driving. Even so, many young drivers want touch-screen interfaces and Smartphone applications available in new cars, extending the reach of e-communication even within the automobile (Deloitte, 2012).

Overall, the role of e-communications in youth travel is still very unclear and requires a great deal of further research.

4. Discussion and Conclusions

Research into the downward trend in youth licensing is still emerging and much of the work is preliminary or indirect. There are likely to be multiple interacting factors behind this trend and the combination of factors will vary between countries and even within cities. However, some initial conclusions can be drawn from this review. Table 3 presents an outline assessment of the likely scale of causal effects based on this review.

Based on the evidence documented so far, changes in life stage and household living arrangements demonstrate the clearest and most consistent impact on changes in youth licensing. Many nations have documented a decline in youth employment, an increase in attending educational institution and a delay in the age of marriage (Australian Bureau of Statistics, 2011c, 2012a, 2012b; Cobb-Clark, 2008; Mitchell, 2006; Office for National Statistics, 2011, 2012; Settersten & Ray, 2010; Taylor et al., 2012). So far three studies have directly linked these factors to a reduction in license-holding; one through a simple comparative survey (Noble, 2005) and two through regression modelling (Delbosc & Currie, 2013; Licaj et al., 2012). No other areas have such clearly documented direct and indirect links.

These changes are likely to be linked to the second set of more likely explanations: changes in the cost of motoring and youth income. However, the evidence of economic changes is not as clear or as consistent. Although in some instances certain motoring costs have increased, such as the rapid increase in insurance premiums for young drivers in the UK (Noble, 2005) or the general increase in petrol prices (Davis et al., 2012), other motoring costs such as car purchase costs are decreasing (Noble, 2005). Even the impact of economic recessions is unclear, as driving continued to decrease in Sweden after the end of a 1990s recession (Berg, 2001) and driving began to decrease before the global financial crisis of 2008 (Davis et al., 2012; Newman & Kenworthy, 2011). It is likely that the trend towards delaying full-time work is reducing the disposable income of young people and that other costs (such as housing) are competing against transport costs; however more research is needed to better understand these complex influences.

The trend in decreased licensing appears to be accompanied by changes in mode use and household location for young adults. Which alternative mode is used varies by nation and city and is sensitive to the nature of the local transport system. There are also clear trends towards increased residential development in inner cities, which should support lifestyles without a car (e.g. Environmental Protection Agency, 2010; Raimond & Milthorpe, 2010). However, it is unclear which is the cause and which is the consequence: do young people prefer to use transit and live in accessible areas, so they choose not to get a license? Or do they choose not to

Table 3. Assessment of causal factors

Explanation	Link to youth license decline	Scale of impact	Rationale for rating
Life stage			
Increasing rate of educational participation	Yes	Medium	Rate of change similar to licensing change
Decreasing full-time employment rates	Yes	Medium	Rate of change similar to licensing change; flow-on effect to affordability
Delaying marriage/children	Yes	Low	Affects only a small share
Living with parents longer	Unclear	Unclear	Studies provide conflicting results
Affordability			
Insurance cost	Yes	Medium to low	Common issue
Cost of petrol	Yes	Low	Not a high share of costs
Cost of car purchase	Unclear	Unclear	Real costs reducing
Disposable income	Unclear	Unclear but possibly high	Complex effect
Recession/economy	Unlikely	Low	Decline occurs outside of recessions
Location and Transport			
Use PT/other modes instead	Yes	Low	Mode shift is small
Moving to inner-city/ accessible areas	Yes	Low	Good evidence but only related to small share of young people
GDL			
Licensing regimes became more strict	Yes	Low	Many cases where decline occurs before/without GDL schemes
Household car access/ driving supervisor	Yes	Low	
Attitudes			
Want to help the environment	Unlikely	Unclear	Little evidence in support
Cars no longer a status symbol	Yes	Low	Attitude differs by country
Too busy/other priorities	Unclear	Unclear	Limited evidence
E-communication			
E-communications replacing face-to-face contact	Unlikely	Unclear	Much further research needed
E-communications suit PT use	Unclear	Unclear	

get a license and then realise that means they must use transit and live in accessible areas? Further research is needed in this area; however, this is also an area where policy and planning can most directly impact youth licensing outcomes. If society chooses to encourage this trend, improvements to the transport system and land-use change should be targeted to facilitate this choice.

The introduction of GDL is likely to have only small effects on youth licensing rates. Licensing rates began to drop in Australia before the introduction of GDL (Raimond & Milthorpe, 2010) and licensing has been dropping in the UK without the introduction of GDL (Directgov, 2012). Further research in this area is needed, especially on the potential impact of supervised driving hours. GDL is considered a barrier to licensing by young people but it is not clear it acts to significantly reduce licensing uptake.

The impact of attitudes and the role of e-communications are still emerging areas for investigation. Indeed the whole search for causal influences is confounded by a lack of longitudinal studies with enough scale to enable reliable measurement of the likely causal factors. In general the reductions in licensing identified are small (under 2% per annum) and the assessment in Table 3 suggests that many factors rather than only one will be influential. This will require a considerable research effort to resolve with any reliable empirical basis.

Even as more becomes known about the causes of youth licensing declines, many important questions remain. It is still unclear to what extent young people are foregoing a license entirely and to what extent they may simply be delaying it until they have sufficient need to drive. Some initial research in the UK suggests that at least for some, as they age they have chosen to forgo a license entirely (Stokes, 2012); this is supported by a cohort analysis from New South Wales which suggests that if a young person does not get a license by their 20s they are unlikely to ever get one (Raimond & Milthorpe, 2010).

While the international decline in youth licensing is a remarkable phenomenon it is important to see it in perspective. The majority of young adults in developed countries still seek and obtain a license and use and buy cars. Nevertheless, a growing minority are delaying or forgoing a driving license, either by choice or due to circumstances. Further research will help us better understand the drivers of these trends but only time will tell us if this new pattern of behaviour becomes something more significant.

References

Australian Bureau of Statistics. (2000). *1998–99 household expenditure survey, Australia* (Report 6530.0).

Australian Bureau of Statistics. (2009). *Home and away: The living arrangements of young people. Australian Social Trends* (Report 4102.0).

Australian Bureau of Statistics. (2010). *Population by age and sex, Victoria* (Report 3101.0).

Australian Bureau of Statistics. (2011a). *2009–10 household expenditure survey, Australia* (Report 6530.0).

Australian Bureau of Statistics. (2011b). *Australian demographic statistics* (Report 3101.0).

Australian Bureau of Statistics. (2011c). *Education and work, Australia (1991–2011)* (Report 6227.0).

Australian Bureau of Statistics. (2011d). *Population by age and sex* (Report 3235.0).

Australian Bureau of Statistics. (2012a). *Australian social trends 2012: Love me do, Commonwealth of Australia* (Report 4102.0).

Australian Bureau of Statistics. (2012b). *Labour force participation, Australia*.

Belden Russonello & Stewart, L. L. C. (2011). *The 2011 Community preference survey: What Americans are looking for when deciding where to live*. Washington, DC: National Association of Realtors.

Berg, H.-Y. (2001). *Understanding subgroups of novice drivers: A basis for increased safety and health*. Linköping: Faculty of Health Sciences, Linköping University.

van den Berg, P., Arentze, T. A., & Timmermans, H. J. P. (2009). Size and composition of ego-centered social networks and their effect on geographic distance and contact frequency. *Transportation Research Record*, *2135*, 1–9.

Bureau of Labor Statistics. (2011). 'School's out'. *Spotlight on statistics*. Retrieved October 10, 2012, from http://www.bls.gov/spotlight/2011/schools_out/

Chamon, M., Mauro, P., & Okawa, Y. (2008). Mass car ownership in the emerging market giants. *Economic Policy, 23*(54), 243–296.

Chapman, L. (2007). Transport and climate change: A review. *Journal of Transport Geography, 15*(5), 354–367.

Chozick, A. (2012, March 22). As young lose interest in cars, G.M. turns to MTV for help. *New York Times*. New York City.

Cobb-Clark, D. A. (2008). Leaving home: What economics has to say about the living arrangements of young Australians. *Australian Economic Review, 41*(2), 160–176.

Davis, B., Dutzik, T., & Baxandall, P. (2012). *Transportation and the new generation: Why young people are driving less and what it means for transportation policy*. Santa Barbara, CA: Frontier Group.

Delbosc, A., & Currie, G. (2012, September 26–28). *Using online discussion forums to study attitudes toward cars and transit among young people in Victoria*. 35th Australasian Transport Research Forum, Perth, Australia.

Delbosc, A., & Currie, G. (2013). *Are changed living arrangements influencing youth driver license decline?* Transportation Research Board 92nd Annual Meeting, Washington, DC.

Deloitte. (2012). *Fourth annual Gen Y automotive survey*.

Department for Environment Food and Rural Affairs. (2002). *Survey of public attitudes to quality of life and the environment — 2001*. London.

Department of Transport. (2009). *Victorian integrated survey of travel and activity (VISTA) 2007*.

Directgov. (2012). *Learners and new drivers and riders*. Retrieved October 4, 2012, from http://www.direct.gov.uk/en/Motoring/LearnerAndNewDrivers/index.htm

Environmental Protection Agency. (2010). *Residential construction trends in America's metropolitan regions*. Development, Community and Environment Devision.

Forward, S., Åretun, A., Engström, I., Nolén, S., & Börjesson, J. (2010). *Young people's attitudes towards acquiring a driving license 2002–2009* [Swedish]. VTI.

Geels, F. W. (2012). A socio-technical analysis of low-carbon transitions: Introducing the multi-level perspective into transport studies. *Journal of Transport Geography*, *24*, 471–482.

Grimal, R., Collett, R., & Madre, J.-L. (2013). Is the stagnation of individual car travel a general phenomenon in France? A time-series analysis by zone of residence and standard of living. *Transport Reviews*. doi:10.1080/01441647.2013.801930

Headicar, P. (2013). The changing spatial distribution of the population in England: Its nature and significance for 'peak car'. *Transport Reviews*. doi:10.1080/01441647.2013.802751

Kalmbach, R., Bernhart, W., Kleimann, P. G., & Hoffman, M. (2011). *Automotive landscape 2025: Opportunities and challenges ahead*. Roland Berger.

KRC Research. (2010). *Millennials & driving; A survey commissioned by Zipcar*.

Kuhnimhof, T., Armoogum, J., Buehler, R., Dargay, J., Denstadli, J. M., & Yamamoto, T. (2012a). Men shape a downward trend in car use among young adults — evidence from six industrialized countries. *Transport Reviews*, *32*(6), 761–779.

Kuhnimhof, T., Buehler, R., & Dargay, J. (2011). A new generation: Travel trends for young Germans and Britons. *Transportation Research Record*, *2230*, 58–67.

Kuhnimhof, T., Buehler, R., Wirtz, M., & Kalinowska, D. (2012b). Travel trends among young adults in Germany: Increasing multimodality and declining car use for men. *Journal of Transport Geography*, *24*, 443–450.

Kuhnimhof, T., Zumkeller, D., & Chlond, B. (2013a). *Who are the 'drivers' of peak car? A decomposition of recent car travel trends for six industrialized countries*. Transportation Research Board 92nd Annual Meeting, Washington, DC.

Kuhnimhof, T., Zumkeller, D., & Chlond, B. (2013b). Who made peak car, and how? A breakdown of trends over four decades in four countries. *Transport Reviews*. doi:10.1080/01441647.2013.801928

Lachman, M. L., & Brett, D. L. (2011). *Generation Y: America's new housing wave*. Washington, DC: Urban Land Institute.

Le Vine, S., Latinopoulos, C., & Polak, J. (2013). A tenuous result: Re-analysis of the link between internet-usage and young adults' driving-licence holding comments on 'recent changes in the age composition of drivers in 15 countries'. *Traffic Injury Prevention*. doi: 10.1080/15389588.2013.793583

Le Vine, S. E., Jones, P. M., & Polak, J. W. (2009). *Has the historical growth in car use come to an end in Great Britain?* 37th European Transport Conference, Leeuwenhorst, The Netherlands.

Licaj, I., Haddak, M., Pochet, P., & Chiron, M. (2012). Individual and contextual socioeconomic disadvantages and car driving between 16 and 24 years of age: A multilevel study in the Rhône Département (France). *Journal of Transport Geography*, *22*, 19–27.

McDonald, N., & Trowbridge, M. (2009). Does the built environment affect when American teens become drivers? Evidence from the 2001 National Household Travel Survey. *Journal of Safety Research*, *40*, 177–183.

Metz, D. (2012). Demographic determinants of daily travel demand. *Transport Policy*, *21*, 20–25.

Metz, D. (2013). Peak car and beyond: The fourth era of travel. *Transport Reviews*. doi:10.1080/01441647.2013.800615

Mitchell, B. A. (2006). The boomerang age from childhood to adulthood: Emergent trends and issues for aging families. *Canadian Studies in Population*, *33*(2), 155–178.

Nantulya, V. M., Sleet, D. A., Reich, M. R., Rosenberg, M., Peden, M., & Waxweiler, R. (2003). Introduction: The global challenge of road traffic injuries. *Injury Control and Safety Promotion*, *10*(1–2), 3–7.

National Travel Survey. (2010), *Full car driving license holders by age and gender* (Table 0201). Department for Transport.

Newman, P., & Kenworthy, J. (2011). 'Peak car use': Understanding the demise of automobile dependence. *World Transport Policy and Practice*, *17*(2), 31–42.

Noble, B. (2005). *Why are some young people choosing not to drive?* European Transport Conference, Strasbourg, France.

Office for National Statistics. (2011). *Young people in the labour market — 2011*. London.

Office for National Statistics. (2012). *Young adults living with parents in the UK — 2011*. London.

Preusser, D. F., & Tison, J. (2007). GDL then and now. *Journal of Safety Research*, *38*, 159–163.

Puentes, R., & Tomer, A. (2008). *The road … less traveled: An analysis of vehicle miles traveled trends in the U.S.* (Metropolitan Infrastructure Initiatives Series). Washington, DC: Brookings Institution.

Raimond, T., & Milthorpe, F. (2010). *Why are young people driving less? Trends in license-holding and travel behaviour.* Australasian Transport Research Forum, Canberra, Australia.

Ruud, A., & Nordbakke, S. (2005). *Decreasing driving licence rates among young people — consequences for local public transport*. European Transport Conference, Strasbourg, France.

Senserrick, T. (2009). Australian graduated driver licensing systems. *Journal of the Australasian College of Road Safety*, *20*(1), 20–28.

Settersten, R. A. J., & Ray, B. (2010). What's going on with young people today? The long and twisting path to adulthood. *The Future of Children: Transition to Adulthood*, *20*(1), 19–41.

Sivak, M., & Schoettle, B. (2011). Recent changes in the age composition of U.S. drivers: Implications for the extent, safety, and environmental consequences of personal transportation. *Traffic Injury Prevention*, *12*, 588–592.

Sivak, M., & Schoettle, B. (2012a). Recent changes in the age composition of drivers in 15 countries. *Traffic Injury Prevention*, *13*, 126–132.

Sivak, M., & Schoettle, B. (2012b). Update: Percentage of young persons with a driver's license continues to drop. *Traffic Injury Prevention*, *13*, 341.

Sivak, M., & Schoettle, B. (2013). Toward understanding the recent large reductions in the proportion of young persons with a driver's license: A response to Le Vine et al. *Traffic Injury Prevention*. doi:10.1080/15389588.2013.793985

Steg, L. (2005). Car use: Lust and must. Instrumental, symbolic and affective motives for car use. *Transportation Research Part A*, *39*, 147–162.

Stokes, G. (2012). *Has car use per person peaked? Age, gender and car use*. Presentation to Transport Statistics Users Group Seminar on Peak Car, London.

Stokes, G., & Hallett, S. (1992). The role of advertising and the car. *Transport Reviews*, *12*(2), 171–183.

Taylor, P., Parker, K., Kochhar, R., Fry, R., Funk, C., Patten, E., & Motel, S. (2012). *Young, underemployed and optimistic: Coming of age, slowly, in a tough economy.* Washington, DC: Pew Research Center.

The Transport Research Centre. (2001). *User manual volume 2 coding frame: A companion document to the VATS94-VATS99 databases*. Melbourne: RMIT University.

Toleman, R., & Rose, G. (2008). Partnerships for progress: Toward sustainable road systems. *Transportation Research Record*, *2067*, 155–163.

VicRoads. (2012). *VicRoads license breakdown by age, 2001–2011*.

van der Waard, J., Jorritsma, P., & Immers, B. (2013). New drivers in mobility; what moves the Dutch in 2012? *Transport Reviews*. doi:10.1080/01441647.2013.801046

Williams, A. F. (2011). Teenagers' licensing decisions and their views of licensing policies: A national survey. *Traffic Injury Prevention*, *12*(4), 312–319.

Zhu, M., Chu, H., & Li, G. (2009). Effects of graduated driver licensing on licensure and traffic injury rates in upstate New York. *Accident Analysis and Prevention*, *41*, 531–535.

Zumkeller, D., Chlond, B., & Manz, W. (2004). Infrastructure development in Germany under stagnating demand conditions: A new paradigm? *Transportation Research Record*, *1864*, 121–128.